

1. Project Methodology

1.1 Research Design

This study employed a **developmental research design**. This design was appropriate because the study focused on designing, developing, and evaluating a software system that addresses the limitations of the existing leave management process at The Lewis College. Rather than merely describing the current process, the study aimed to develop a functional system called **LeaveSync: A Leave Management System with Device Fingerprinting and Digital Signature-Based Approval for The Lewis College**.

The study used qualitative data gathering to understand the existing leave management process and identify the requirements of the proposed system. Information was gathered through interviews, observation, and review of existing procedures and documents related to leave filing and approval. The gathered information served as the basis for designing the features, workflow, security mechanisms, and approval process of LeaveSync.

1.2 Software Development Methodology

The development of LeaveSync followed an **Agile iterative-incremental development methodology**. Agile was selected because the proposed system contains several interconnected features, particularly authentication, multi-factor authentication, device fingerprinting, leave validation, approval routing, and digital signature verification. Developing these features incrementally allowed the researchers to implement, test, evaluate, and improve individual system components before integrating them into the complete system.

The development process consisted of the following recurring activities:

Requirements Gathering and Analysis. The researchers identified the functional and security requirements of the system based on the existing leave management process. The requirements included user authentication, multi-factor authentication, device verification, leave request submission, leave balance validation, approval routing, and digital signature-based approval.

System Design. The identified requirements were translated into system designs, database structures, process models, and interface designs. BPMN and UML Activity Diagrams were used to represent the existing and proposed processes.

Implementation. The researchers developed the LeaveSync web application using PHP for the backend, MySQL for data storage, and HTML, CSS, and JavaScript for the frontend. Security-related functions were implemented using appropriate authentication, device fingerprinting, and cryptographic technologies.

Testing and Evaluation. Individual system functions were tested to determine whether they operated according to the specified requirements. The researchers also performed functional testing of the complete workflow, including login, multi-factor authentication, trusted-device verification, leave submission, supervisor approval, HR approval, and leave balance updating.

Review and Refinement. Errors and inconsistencies identified during testing were corrected and the affected components were refined. The development cycle was repeated until the major system functions operated according to the identified requirements.

The iterative process allowed the researchers to progressively develop the system through the following major stages:

Authentication → Multi-Factor Authentication → Device Fingerprinting → Leave Request → Leave Validation → Approval Routing → Digital Signature Verification → Leave Balance Update → Audit Logging

1.3 Data Gathering Procedures

The researchers gathered information about the existing leave management process through the following procedures:

Interview. Interviews were conducted with personnel involved in the leave management process to determine how leave requests are currently submitted, reviewed, approved, and recorded. The interviews also helped identify difficulties and possible improvements in the existing process.

Observation. The researchers observed the existing leave request process to understand the actual sequence of activities performed by employees, supervisors, and HR personnel. Particular attention was given to manual submission, notification, approval, encoding, and leave balance updating.

Document Review. Existing leave-related forms, procedures, and records were reviewed to identify the information required in a leave request and the manner in which approved or rejected requests are recorded.

The information gathered from these procedures was used as the basis for analyzing the existing system and designing the proposed LeaveSync process.

1.4 System Development Tools and Technologies

The following technologies and tools were used in the development of LeaveSync:

Category	Tool / Technology
Backend	PHP
Database	MySQL
Frontend	HTML5, CSS3, JavaScript
Authentication	Multi-Factor Authentication (MFA)
Device Security	Device Fingerprinting
Digital Signature	WebAuthn/FIDO2 Passkeys and Cryptographic Verification
Cryptography	OpenSSL, RSA, SHA-256
Version Control	Git and GitHub
Diagramming	PlantUML
Development Environment	Visual Studio Code

These technologies were selected to support the functional requirements and security features of the LeaveSync system.

2. Conceptual Framework

The conceptual framework of the study follows the **Input-Process-Output (IPO) Model**. The model illustrates the relationship between the information required by LeaveSync, the processes performed by the system, and the outputs generated after the processing of a leave request.

2.1 Input

The inputs of the LeaveSync system include the information and conditions required to process a leave request. These include:

- Employee account information;
- Employee authentication and multi-factor authentication credentials;
- Trusted-device information generated through device fingerprinting;
- Selected leave type;
- Leave start date;
- Leave end date;
- Leave reason;

- Employee leave balance;
- Approval decisions from authorized approvers; and
- Digital signature or WebAuthn passkey assertion from the authorized approver.

Information regarding the existing manual leave process was also considered during system analysis and served as a basis for identifying the requirements and weaknesses that the proposed system needed to address.

2.2 Process

The process begins when an employee accesses LeaveSync and authenticates their account. The system verifies the user's authentication status, multi-factor authentication, and trusted-device information.

After successful authentication, the employee enters the required leave information. LeaveSync validates the request by checking the validity of the selected leave type, date range, number of weekdays, existing overlapping requests, and available leave balance.

If the request passes validation, the system creates the leave request and temporarily reserves the required number of leave days. The request is then routed to the appropriate approver.

For regular employees, the request is first submitted to the assigned supervisor. Once approved by the supervisor, the request is forwarded to HR for final approval. Requests submitted by managers are routed directly to HR.

Before an authorized approver can approve a request, the system requires verification of the approver's registered passkey through WebAuthn. This provides cryptographic evidence that the approval was performed by the authorized account holder.

If the request is approved, the reserved leave days are transferred to used leave days. If the request is rejected, the reserved days are returned to the employee's available leave balance. The system also records the actions performed during the process in the audit log.

2.3 Output

The outputs of the LeaveSync process include:

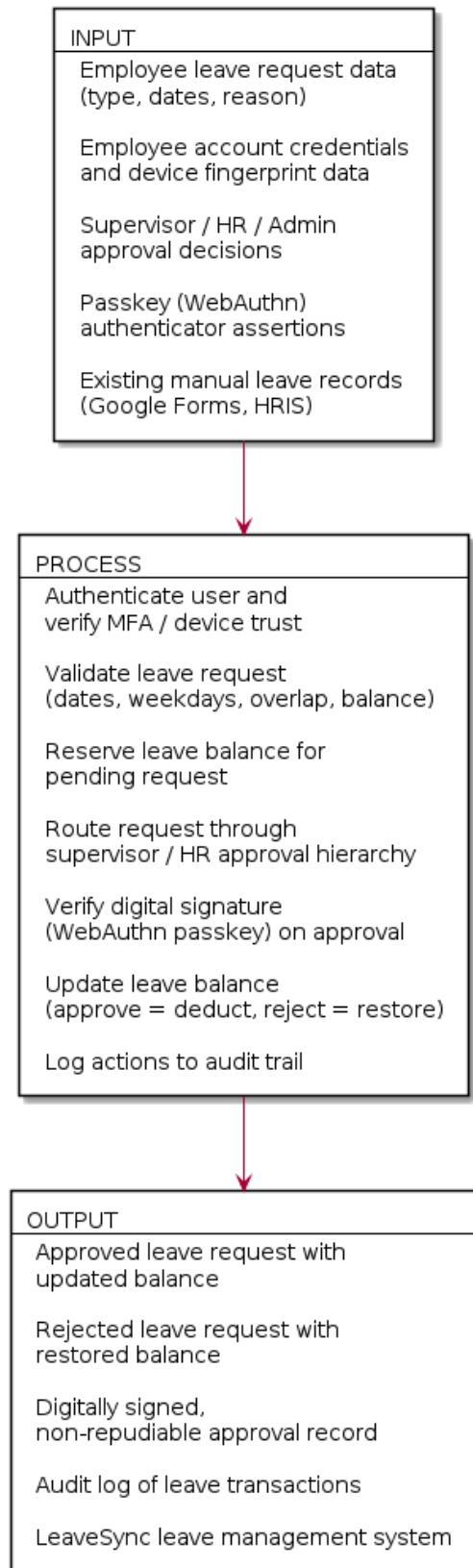
- An approved leave request with updated leave balance;
- A rejected leave request with the reserved leave days restored;
- A digitally verified approval or rejection record;
- A record of the approver and time of the action;
- An updated leave request status; and
- An audit trail of important system activities.

Through these outputs, LeaveSync provides a centralized and more secure leave management process that reduces manual encoding, improves approval tracking, and strengthens the authenticity and accountability of approval actions.

2.4 Conceptual Framework Diagram

The conceptual framework can be represented using the following IPO structure:

Conceptual Framework (Input-Process-Output Model)



3. Analysis of the Existing System

3.1 Existing Process

The existing leave management process of The Lewis College uses a **manual workflow involving Google Forms, email or Google account notifications, and the Human Resource Information System (HRIS)**.

The employee begins the process by submitting a leave request through Google Forms. The submitted information is received through the appropriate notification channel by the supervisor. The supervisor reviews the request and decides whether to approve or reject it.

After making a decision, the supervisor manually records the result in the HRIS under the employee's account. If the request is approved, the corresponding leave days are deducted from the employee's leave balance. If the request is rejected, the employee's leave balance remains unchanged.

The process therefore requires the supervisor to perform an additional manual encoding step after reviewing the request.

3.2 Objective of the Existing Process

The objective of the existing process is to allow employees to submit leave requests and enable the responsible supervisor to review the request and record the approval or rejection in the HRIS.

3.3 Participants and Systems

The existing process involves the following participants and systems:

- **Employee** – submits the leave request;
- **Google Forms** – receives and stores the submitted leave request;
- **Supervisor** – receives, reviews, and decides on the leave request;
- **Google Account/Email Notification** – serves as the notification channel for the supervisor; and
- **HRIS** – stores the final leave result and updates the employee's leave balance.

3.4 Inputs of the Existing Process

The existing process requires the following information:

- Employee identity;

- Leave type;
- Leave start date;
- Leave end date; and
- Leave reason.

3.5 Outputs of the Existing Process

The existing process produces either:

- An approved leave request recorded in the HRIS with the corresponding leave balance deduction; or
- A rejected leave request recorded in the HRIS without a leave balance deduction.

3.6 Business Rules of the Existing Process

The existing process follows these basic rules:

1. The employee submits the leave request using Google Forms.
2. The submitted request is received through the supervisor's notification channel.
3. The supervisor reviews the submitted leave information.
4. The supervisor either approves or rejects the request.
5. The supervisor manually enters the decision into the HRIS.
6. An approved request results in a deduction from the employee's leave balance.
7. A rejected request does not reduce the employee's leave balance.

3.7 Weaknesses of the Existing System

The analysis of the existing process identified several weaknesses.

Lack of device and identity verification. The process does not provide a dedicated mechanism for verifying whether the person submitting the request is using an authorized device or account.

Lack of cryptographic approval verification. Approval is represented by a manual entry into the HRIS. There is no cryptographic signature that can provide strong evidence that the approval was performed by the authorized approver.

Manual encoding. The supervisor needs to manually transfer the decision from the submitted request into the HRIS. This additional step may result in encoding errors, omissions, or delays.

Limited centralized auditability. The existing process does not provide a dedicated consolidated audit trail showing all important actions, including who performed an action, what action was performed, and when it occurred.

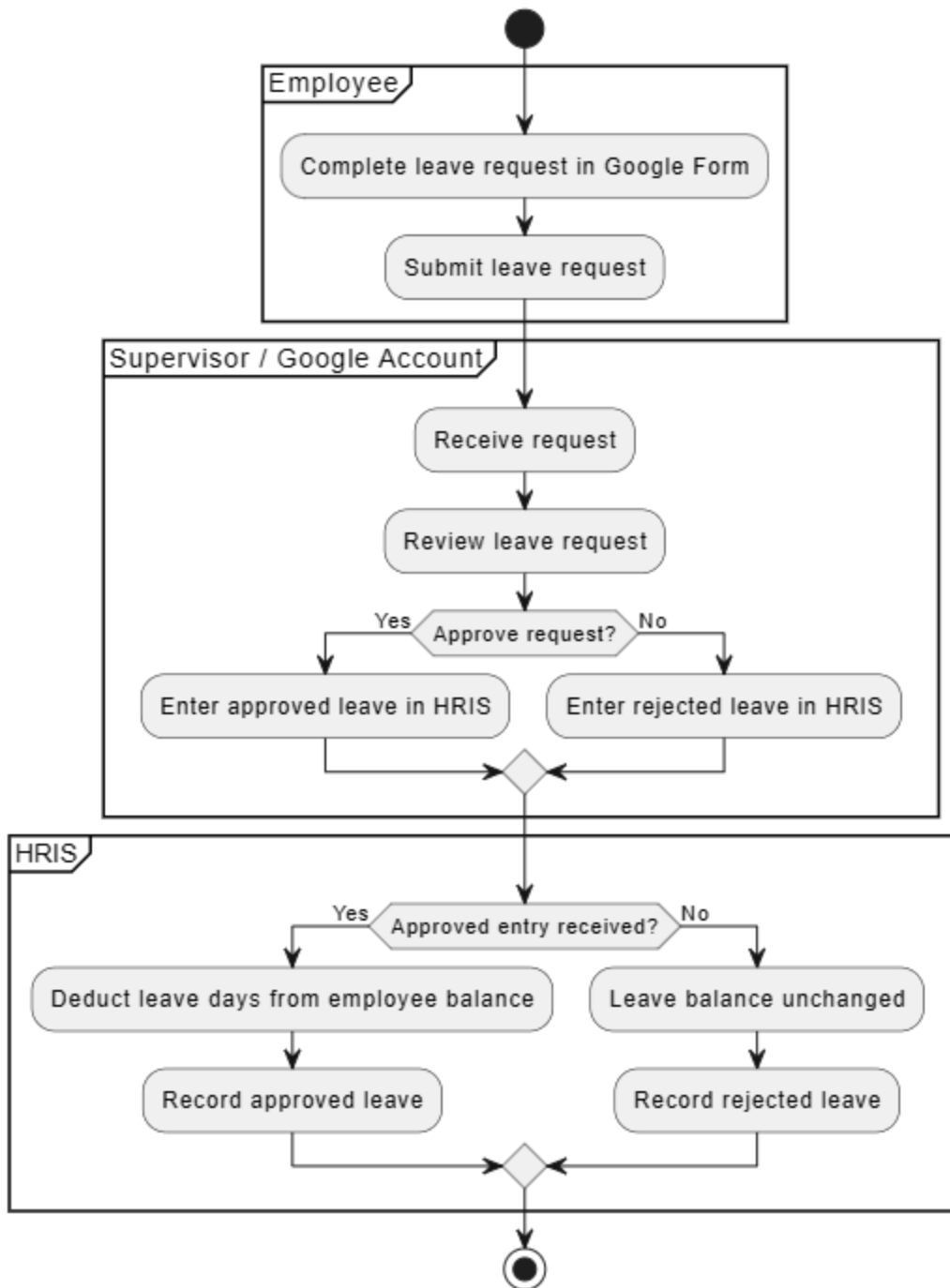
No automatic overlap prevention. The process does not automatically prevent an employee from submitting overlapping leave requests. Verification depends on manual checking.

Delayed balance validation. Leave balance checking and updating depend on the manual processing of the request in the HRIS, which can introduce delays between submission, approval, and balance updating.

3.8 BPMN Diagram of the Existing System

The following BPMN-style diagram represents the existing manual leave request and approval process.

Current Leave Approval BPMN-Style Process



BPMN Interpretation

The BPMN-style representation can be divided into four major lanes: **Employee**, **Google Forms/Notification Channel**, **Supervisor**, and **HRIS**.

The **Employee lane** begins the process. The employee completes the required information and submits the leave request using Google Forms.

The request then moves to the **Google Forms/Notification Channel**, where the submitted information is received and the supervisor is notified.

The **Supervisor lane** represents the review and decision-making stage. The supervisor examines the submitted leave request and decides whether to approve or reject it. This decision is represented by an exclusive gateway because only one of the two outcomes can occur.

If the request is **approved**, the supervisor manually enters the approval into the HRIS. The HRIS then records the approved leave and deducts the corresponding number of days from the employee's leave balance.

If the request is **rejected**, the supervisor manually enters the rejection into the HRIS. The HRIS records the rejection and does not deduct leave days from the employee's balance.

The process ends after the HRIS records the supervisor's decision and performs the appropriate leave balance action.

4. New System Process Using UML Activity Diagram

4.1 Process Name

Submit and Approve Leave Request in LeaveSync

4.2 Objective

The objective of the proposed process is to allow employees to submit leave requests through LeaveSync and enable authorized supervisors and HR personnel to review and approve requests through a controlled, automated, and digitally verified workflow.

The system automatically performs leave validation, overlap checking, leave balance reservation, approval routing, digital signature verification, and leave balance updating.

4.3 Inputs

The proposed LeaveSync process requires the following inputs:

- Employee account;
 - Multi-factor authentication credentials;
 - Trusted-device information;
 - Selected leave type;
 - Leave start date;
 - Leave end date;
 - Leave reason;
 - Available leave balance; and
 - Digital signature/passkey assertion from an authorized approver.
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4.4 Outputs

The proposed process produces one of the following results:

- A pending leave request with reserved leave balance;
 - An approved leave request with the corresponding days transferred to used leave;
 - A rejected leave request with the reserved days returned to the available balance;
 - A digitally verified approval or rejection record; and
 - An audit record of the actions performed throughout the process.
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4.5 System Rules

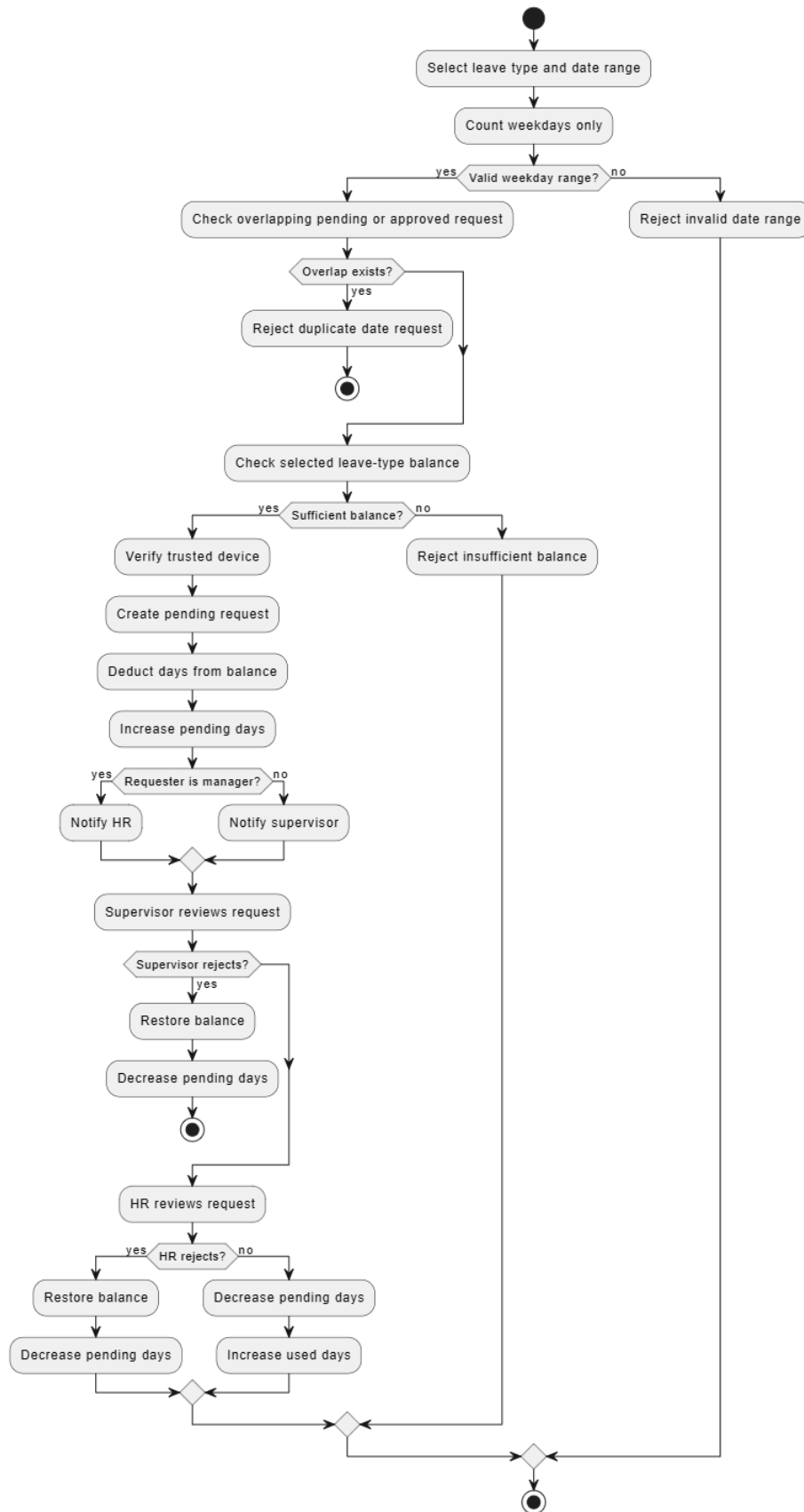
LeaveSync follows the following rules:

1. The selected leave type must be available to the employee.
2. The end date must not be earlier than the start date.
3. Saturdays and Sundays are not counted as leave days.
4. A date range containing no weekdays cannot be submitted.
5. An employee cannot submit a request that overlaps an existing pending or approved request.
6. The employee must have sufficient available leave balance.
7. The request must originate from a registered and trusted device.
8. Regular employee requests are routed first to the assigned supervisor and then to HR.
9. Manager requests are routed directly to HR.
10. Approval actions require verification of the authorized approver's passkey.

11. An approved request transfers reserved days to used leave.
 12. A rejected request returns the reserved days to the employee's available balance.
 13. Important actions are recorded in the system audit trail.
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4.6 UML Activity Diagram of the New System

Leave Request and Approval Activity



4.7 UML Activity Diagram Explanation

Initial Node

The solid black circle represents the starting point of the activity. The process begins when the employee accesses the New Request page in LeaveSync.

Enter Leave Request Information

The employee selects the appropriate leave type and provides the leave start date, end date, and reason for the leave. These values are used by the system during the validation process.

Count Weekdays

LeaveSync calculates the number of leave days covered by the requested date range. Saturdays and Sundays are excluded from the calculation because they are not counted as leave days.

Validate Date Range

The system checks whether the end date is earlier than the start date and whether the requested period contains at least one weekday.

If the date range is invalid, the system displays an error and returns the employee to the request form.

Check for Overlapping Requests

The system checks whether the employee already has a pending or approved leave request that overlaps the requested dates.

If an overlapping request exists, the system prevents the new request from being submitted and displays an appropriate error message.

Check Leave Balance

The system checks the employee's available balance for the selected leave type.

If the employee does not have enough available leave days, the request cannot proceed.

Verify Trusted Device

Before the request is created, LeaveSync verifies whether the employee is using a registered and trusted device. The system uses device fingerprinting to identify the device associated with the employee's account.

If the device is not trusted, the system prevents the submission from proceeding.

Create Pending Request and Reserve Leave Days

After all validations are successfully completed, LeaveSync creates the leave request with a **Pending** status.

The required leave days are immediately reserved by reducing the available balance and adding the corresponding number of days to pending leave. This prevents the same available days from being used by another pending request.

Determine Approval Route

The system determines the appropriate approval route based on the employee's role.

For a regular employee, the request is first routed to the assigned supervisor. After supervisor approval, the request proceeds to HR for final approval.

For a manager, the request bypasses the supervisor and is routed directly to HR.

Verify Approver's Passkey

Before an authorized approver can approve the request, LeaveSync requires the approver to authenticate using a registered WebAuthn/FIDO2 passkey.

The system verifies the passkey assertion and confirms that the approval action is associated with the authorized approver.

If the passkey verification fails, the approval action is not completed.

Supervisor Decision

For regular employee requests, the supervisor reviews the leave request and selects either **Approve** or **Reject**.

If the supervisor rejects the request, the request is marked as rejected and the previously reserved leave days are returned to the employee's available balance.

If the supervisor approves the request, the request proceeds to the HR approval stage.

HR Decision

HR performs the final approval decision. Before the decision is committed, HR must also complete passkey verification.

If HR rejects the request, the reserved leave days are returned to the employee's available balance.

If HR approves the request, the reserved leave days are transferred from pending leave to used leave.

Balance Transfer

The system does not deduct the leave days twice. The days were already reserved when the request was created.

Therefore, when a request is approved, LeaveSync simply transfers the reserved days from **pending leave** to **used leave**.

When a request is rejected, the reserved days are returned to **available leave balance**.

Audit Logging

After the approval or rejection process is completed, LeaveSync records the important actions in the audit trail. The audit record provides information about the action performed, the user who performed it, and the corresponding time of the action.

Activity Final Node

The activity reaches the final node after the request has been approved or rejected, the appropriate leave balance has been updated, and the corresponding activity has been recorded in the audit trail.

The final status is then available to the appropriate employee and authorized personnel.

4.8 Summary of the Proposed Process

The proposed LeaveSync process improves the existing leave management workflow by replacing several manual activities with automated system processes. Instead of submitting a request through a separate form and requiring manual encoding into the HRIS, employees submit their requests directly through LeaveSync.

The system automatically validates the request, checks for overlapping requests, verifies available leave balance, confirms the trusted device, reserves the requested leave days, and routes the request to the appropriate approver.

Approval is strengthened through WebAuthn passkey verification, providing cryptographic verification of the approver's action. Once the final decision is made, LeaveSync automatically updates the employee's leave balance and records the relevant actions in the audit trail.

Through this process, LeaveSync provides a more centralized, automated, and secure approach to leave management while addressing the major weaknesses identified in the existing manual process.